

OligoTox Open Data Challenge — Phase 2 Narrative Document

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Submission target: Single PDF, ≤12 pages, 8.5"×11", ≥1" margins, ≥11pt Arial, line spacing ≥1.0

Challenge: NIH NCATS Oligonucleotide Toxicity (OligoTox) Open Data Challenge — Phase 2 (Data Generation)

STATUS: DRAFT / PLACEHOLDER. This document is structured to match the six required narrative sections exactly (challenge announcement, "Phase 2 Data Generation Phase," items 1–6). Sections 1–2 and 4–6 currently describe *planned* work because no wet-lab data has been generated yet. Per the requirement, Phase 2 is judged on "the demonstrated dataset as it is submitted" — this document must be rewritten in the past tense with real numbers, real distributions, and real findings before submission. Do not submit in its current form.

1. Executive Summary of the Dataset(s) Generated, and Positive/Negative Controls Included

[PLACEHOLDER — fill in once Batch A–E data collection is complete or substantially underway.]

Target content once real data exists:

- Final oligo count actually synthesized and screened (library design targets ~2,800; report actual N)
- Which toxicity domains were actually covered (hepatotoxicity, nephrotoxicity, immunotoxicity,

complement activation, coagulopathy, thrombocytopenia — per the challenge's priority list)

- Which in vitro human systems were actually run (of the 5 proposed: PHH, kidney proximal

tubule organoids, PBMC co-culture, platelet-rich plasma, liver-kidney MPS)

- The actual positive/negative controls included and their observed responses (see Appendix A

of the design plan for the proposed control list — CpG ODN 2006, Poly-G ODN, ISIS 104838, Nusinersen, Mipomersen, scrambled PS-ASO, etc.) — report whether each behaved as expected

2. Summary of the Main Findings and Conclusions

[PLACEHOLDER — requires real assay results.] This section must report actual outcomes, e.g.:

- Which chemistries/motifs showed the strongest toxicity signal in which system
- Whether control compounds validated the assay panel (Z'-factor, CV, expected direction of effect)
- Any unexpected findings or negative results (both are valuable for an open dataset)
- Summary statistics (e.g., % of library flagged toxic per endpoint, dose-response quality metrics)

3. Description of How Data Were Produced

This section can largely be finalized now, since it describes process rather than results.

Experimental design: Tiered screening across five human in vitro systems (see 02_methodology_document.md for full protocol detail). Library of oligonucleotides spanning

multiple backbone chemistries (PS, PO, PMO, LNA-mix, PNA, 2'-F-ANA, morpholino), sugar modifications (DNA, 2'-OMe, 2'-F, LNA, mixed), and structural classes (gapmers, siRNA, aptamers, splice-switching oligos, GalNAc/lipid conjugates), plus systematic CpG-content, GC-content, and length series, alongside known positive/negative controls.

Data acquisition: Automated liquid handling (acoustic dispensing for compound transfer, programmable handlers for cell/reagent dispensing) with plate-reader-based endpoint capture (luminescence, fluorescence, absorbance, multiplex immunoassay) and, for a subset, RNA-seq and proteomics.

Computational processing: Raw plate reader output → normalization to vehicle control → per-plate QC (Z'-factor ≥ 0.5 , control CV $\leq 15\%$) → dose-response curve fitting (4-parameter logistic) to derive IC₅₀/EC₅₀, Emax, Hill coefficient, NOEC, and AUDRC → batch-level drift correction via repeated anchor compounds.

[Once real data exists, replace the above generic description with actual batch counts, actual QC pass/fail rates, and any deviations from the planned protocol.]

4. Description of How Indicators and Predictor Variables Were Measured, Their Distribution, and the Distribution of Predictor Variables Amongst Tested Oligos

[PLACEHOLDER — requires the actual assembled dataset.]

Indicators (toxicity readouts) and predictor variables (sequence/chemistry features) are defined

in dataset/data_dictionary_and_schema.md. Once real data exists, this section must report:

- Distribution of each indicator across the tested library (e.g., histogram/summary stats of viability, cytokine levels, IC50 values, by assay system and endpoint)
- Distribution of predictor variables amongst tested oligos (e.g., how many oligos per backbone

class, sugar modification, length bucket, GC-content bin, CpG count) — i.e., does the library actually achieve the chemical diversity it was designed for, or are there gaps/imbbalances

- Missingness/coverage: which oligo × endpoint combinations have data vs. do not

5. Discussion of How the Results Address a Gap in the Publicly Available Data

Argument that can be drafted now, to be tightened once real scale/composition is known:

- Existing public oligo toxicity data is dominated by small (<200 compound), single-endpoint,

animal-derived, PS-ASO-heavy datasets (see literature review in Phase 1 submission)

- This dataset's contribution: human in vitro origin, multi-endpoint (hepatotox, nephrotox, immunotox, complement, coagulopathy, thrombocytopenia in one coordinated panel), and chemically diverse (multiple backbone/sugar/conjugate classes) coverage

- *[Once real data exists: quantify the actual gap closed — e.g., "adds N chemically-diverse

human-system data points, a Xx increase over the largest comparable public dataset."]*

6. Discussion of How the Data Could Be Used to Develop a Predictive Model

- Structured, multi-endpoint labels support both single-task and multi-task model training
- Chemistry + sequence features (documented in the data dictionary) support both interpretable

feature-based models (e.g., gradient-boosted trees with SHAP) and sequence foundation-model

fine-tuning

- Dose-response structure (not just binary tox/non-tox) supports regression and uncertainty-aware

modeling, and benchmark splits (random, chemistry-stratified, prospective) support rigorous out-of-distribution evaluation

- *[Once real data exists: report actual baseline model performance on the real dataset — not

simulated numbers — if any modeling has been done as part of the submission.]*

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